SWINBURNE UNIVERSITY OF TECHNOLOGY

Exploratory Data Analysis Report
Face Recognition Dataset

Facial Recognition with Emotion and Liveness
COS30082 Applied Machine Learning, Group Project

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Source notebook: Face Recognition (Data Loading and EDA)

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Data: Google Drive folder “Face Recognition” (classification_data, verification_data)

1. Summary

This report describes the face dataset that our team will use to train the face recognition model. Before training any model, we checked *what is in the data*: how many people and images there are, whether the images are clean, whether the data is balanced, and whether the test data accidentally contains images the model has already seen. All numbers come from running the project notebook on the real dataset.

Question	Answer	Verdict
How big is the dataset?	4,000 people; 380,699 training, 8,003 validation, 8,000 test and 69,101 verification images	Large enough
Are the files clean?	No corrupted files. All images are 64x64 JPEG, RGB	Good
Is it balanced?	15 to 249 images per person (average 95). Gini 0.20	Reasonably balanced
Is image quality good?	About 7% are dark, 1% greyscale, almost none blurry (see note on the blur threshold)	Acceptable
Are faces varied?	Mostly frontal (63%), 37% turned; many expressions and lighting conditions	Good
Is there data leakage?	Classification splits: none. Verification images: 73 of 6,910 checked are also in training	Needs action

Main message
The classification data (train / val / test) is clean and ready to use. The main problem is **leakage** in the verification folder: about 1% of the checked images are exact copies of training images. These must be removed before we report any verification score, otherwise the result will look better than it really is.

Important limit of this run
The quality scan opened a random **10%** of the images (39,670 of 396,699 classification images and 6,910 of 69,101 verification images), because reading files straight from Google Drive is slow. Percentages are reliable estimates; counts of leaked images are **minimum** values. A full scan (using a zip file) will give exact numbers.

2. What data we have and how we checked it

The shared folder has two parts. **classification_data** is split by the course into *train_data*, *val_data* and *test_data*; each contains one sub-folder per person. The same 4,000 people appear in all three splits (this is called a *closed-set* task: the model learns to name people it has seen). **verification_data** is one flat folder of images with no names and no sub-folders. It is meant for the verification task: deciding whether two photos show the same person.

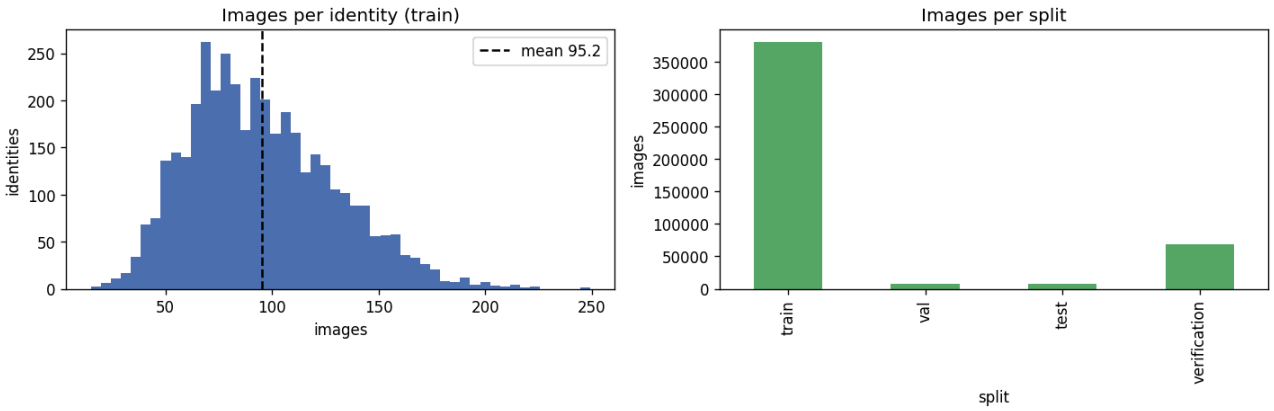


Figure 1. Images per person in training (left) and images per data group (right).

How the checks work

Check	Plain-language meaning	How it is measured
Size / format	Are all images the same shape and file type?	Width, height, JPEG / RGB
Sharpness	Is the picture blurry?	Variance of the Laplacian (an edge detector). Low = blurry
Brightness / contrast	Too dark, too bright, or washed out?	Average and spread of grey levels (0 to 255)
Duplicates	Is the same picture stored twice?	MD5 hash = identical file; perceptual hash = near copy
Pose / expression	Do faces look in different directions and show different expressions?	Google MediaPipe face landmarks on a random sample of 3,000 training images
Leakage	Does the test data contain training pictures?	Compare hashes between the splits

3. Classification data

3.1 Size and balance

Split	Identities	Images	Images per person (min / mean / max)
Training	4,000	380,699	15 / 95.2 / 249
Validation	4,000	8,003	2 / 2.0 / 4
Test	4,000	8,000	2 / 2.0 / 2

Every person has at least 15 training images, so there are no “weak” identities with too few examples to learn from. The histogram in Figure 1 shows a smooth bell shape with a mild tail to the right: a few people have up to 249 images, but the most common range is 60 to 120. The Gini coefficient is 0.20 (0 means perfectly even, 1 means very uneven), and the 10% of people with the most images hold 17% of all images, close to the 10% of a perfectly even dataset. We therefore do not need to re-balance the data, although a class-weighted loss or balanced sampling is still a cheap safeguard.

> **What this means for training**
> Validation has only 2 images per person and test exactly 2. Accuracy on 8,000 test images is reliable overall, but results for a single person rest on just 2 photos.

3.2 Image properties and quality

Every scanned image is a 64x64 JPEG in RGB colour. Because all images already have the same size, no resizing is needed for the input of a 64x64 network; for networks that expect larger inputs (for example 112x112) the images must be up-scaled, which cannot add detail.

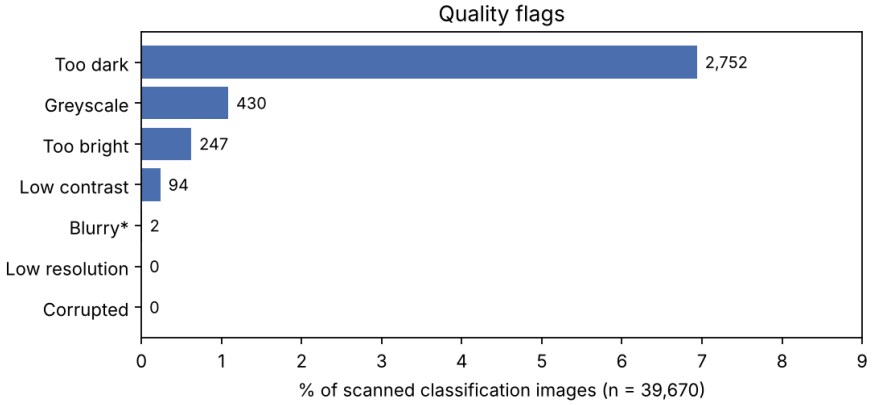


Figure 2. Share of scanned images that trigger each quality flag. *Blurry uses the starting threshold of 50 (see Section 3.3).

Flag	Train	Val	Test	Meaning
Corrupted	0	0	0	File cannot be opened
Low resolution (<32 px)	0	0	0	Image too small
Too dark (mean <60)	2,646	56	50	About 7% of images in each split
Too bright (mean >190)	236	4	7	Under 1%
Greyscale	411	6	13	Black-and-white photos stored as RGB, about 1%
Low contrast	90	4	0	Flat, washed-out images
Blurry (threshold 50)	2	0	0	Almost none at this threshold
Images scanned	38,067	787	816	10% random sample

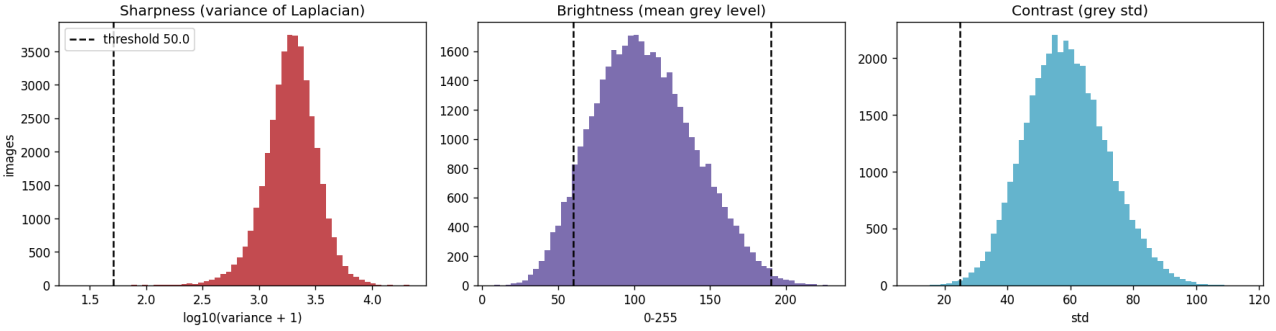


Figure 3. Distribution of sharpness, brightness and contrast. Dashed lines are the thresholds used to raise a flag.

The three histograms are smooth, single-peaked and similar in shape for all splits, which is what we hope to see in a dataset collected from many sources. The brightness curve has a small tail on the dark side, which is the 7% flagged as too dark.

3.3 Looking at the extreme images

Numbers cannot tell us whether a “dark” image is still a usable face, so we looked at the extremes.

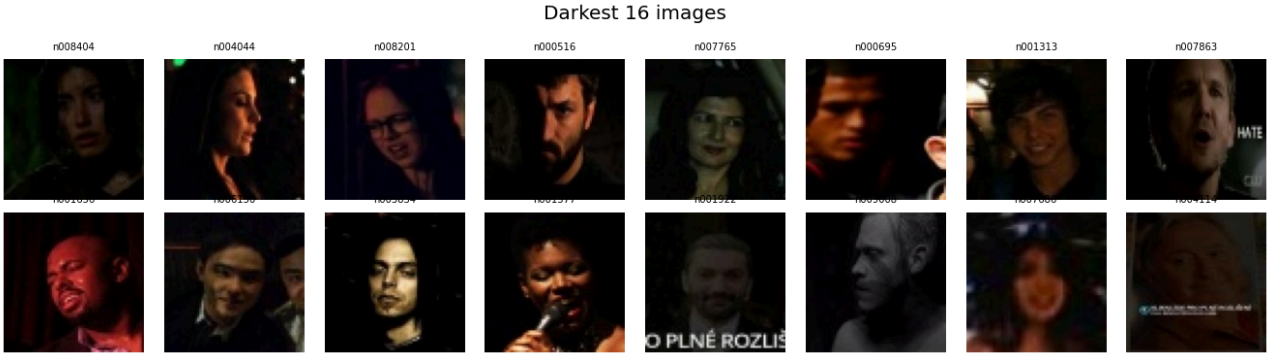


Figure 4. The 16 darkest images. Most are night or concert scenes; faces remain visible.

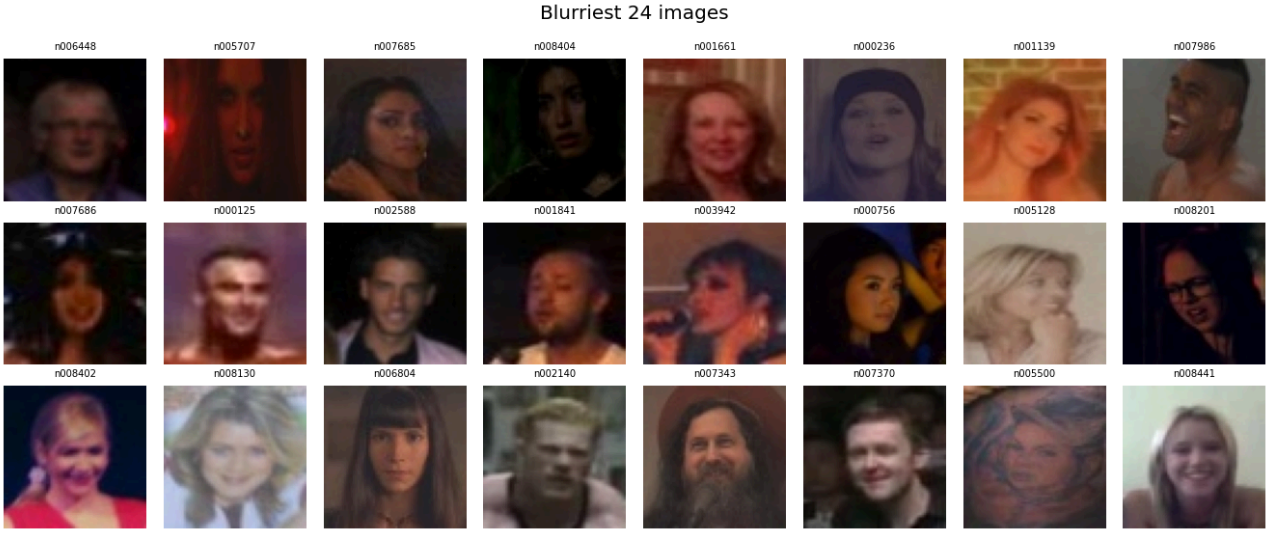


Figure 5. The 24 blurriest images.

Finding: the blur threshold is too low
The sharpness values of normal images sit around 2,000 (the peak in Figure 3 is at about 10^3.3), while the starting threshold is 50. As a result only 2 images were flagged, even though Figure 5 shows images that are visibly soft or out of focus. The 1st percentile of sharpness is 452, so a threshold between roughly 450 and 700 would flag the worst 1 to 5% of images. We recommend choosing it by inspecting grids like Figure 5, and removing blurry images from **training only**.

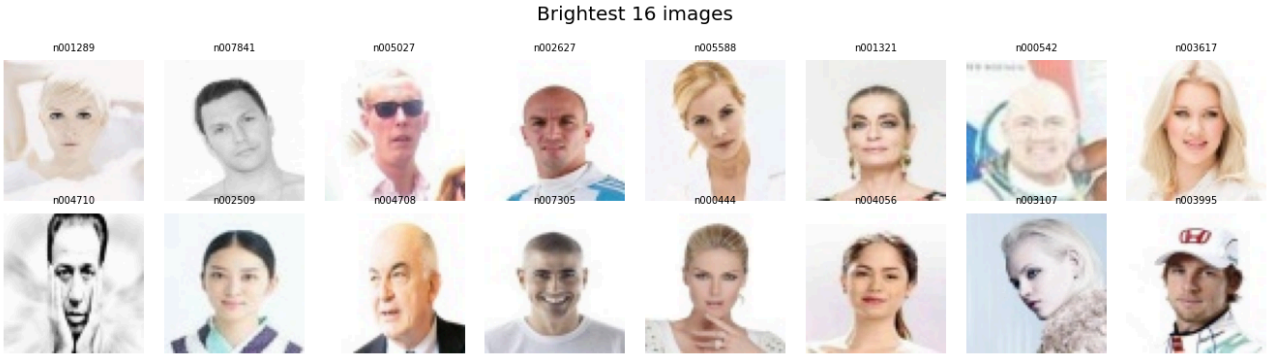


Figure 6. The 16 brightest images (over-exposed or white background).

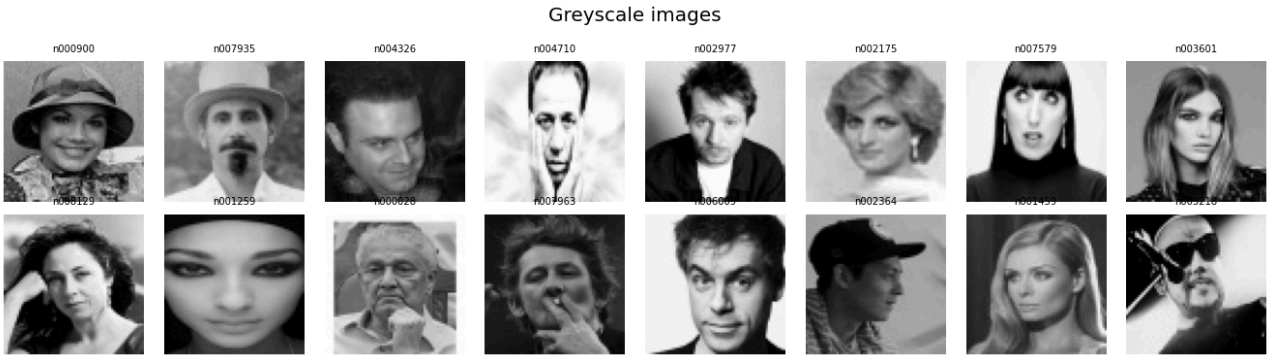


Figure 7. Greyscale images (about 1% of the data). They are normal faces; they could be kept, or colour-jitter augmentation can be used so the model does not depend on colour.

3.4 Lighting, pose and expression

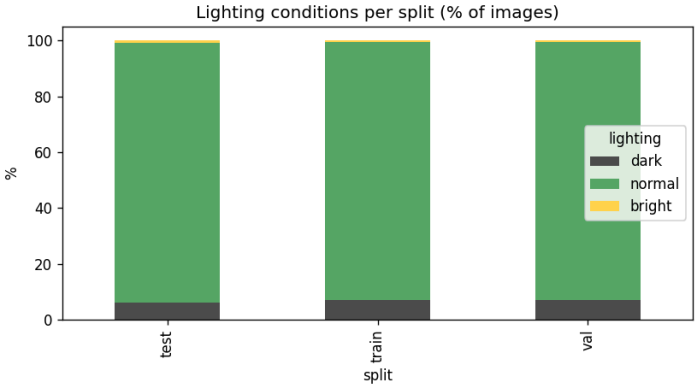


Figure 8. Lighting conditions in each split.

The share of dark, normal and bright images is almost identical in train, validation and test (about 7% dark, 92% normal, under 1% bright). This is good news: the test set is not harder or easier than the training set because of lighting.

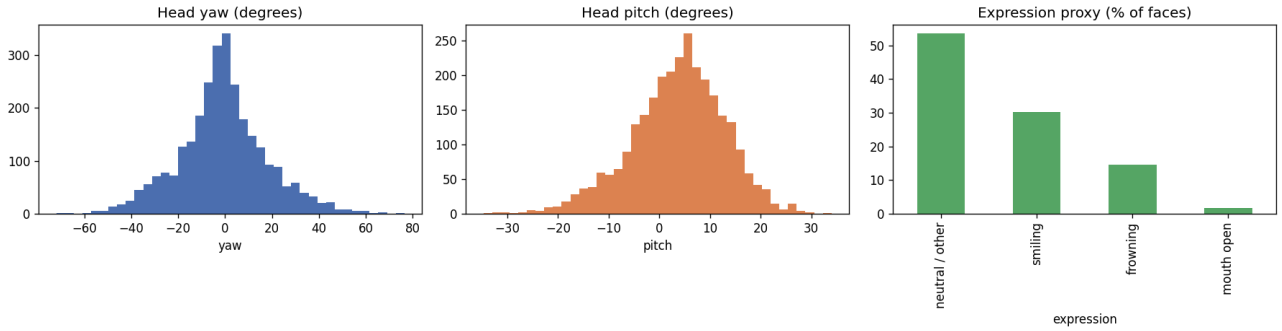


Figure 9. Estimated head turn (yaw), head tilt (pitch) and expression on a random sample of 3,000 training images.

Head pose (yaw)	Share	Expression proxy	Share
Frontal (under 15°)	62.9%	Neutral / other	53.5%
Semi-profile (15 to 35°)	29.3%	Smiling	30.2%
Profile (over 35°)	7.8%	Frowning	14.6%
		Mouth open	1.7%

Nearly two thirds of the faces look at the camera, but more than one third are turned by 15° or more, and 8% are close to a side view. This variety is useful because the live camera system will not always receive a

perfectly frontal face. Expression and pose are *estimates* from facial landmarks on small 64x64 images, so they are hints, not exact labels.

3.5 Who is in the dataset?

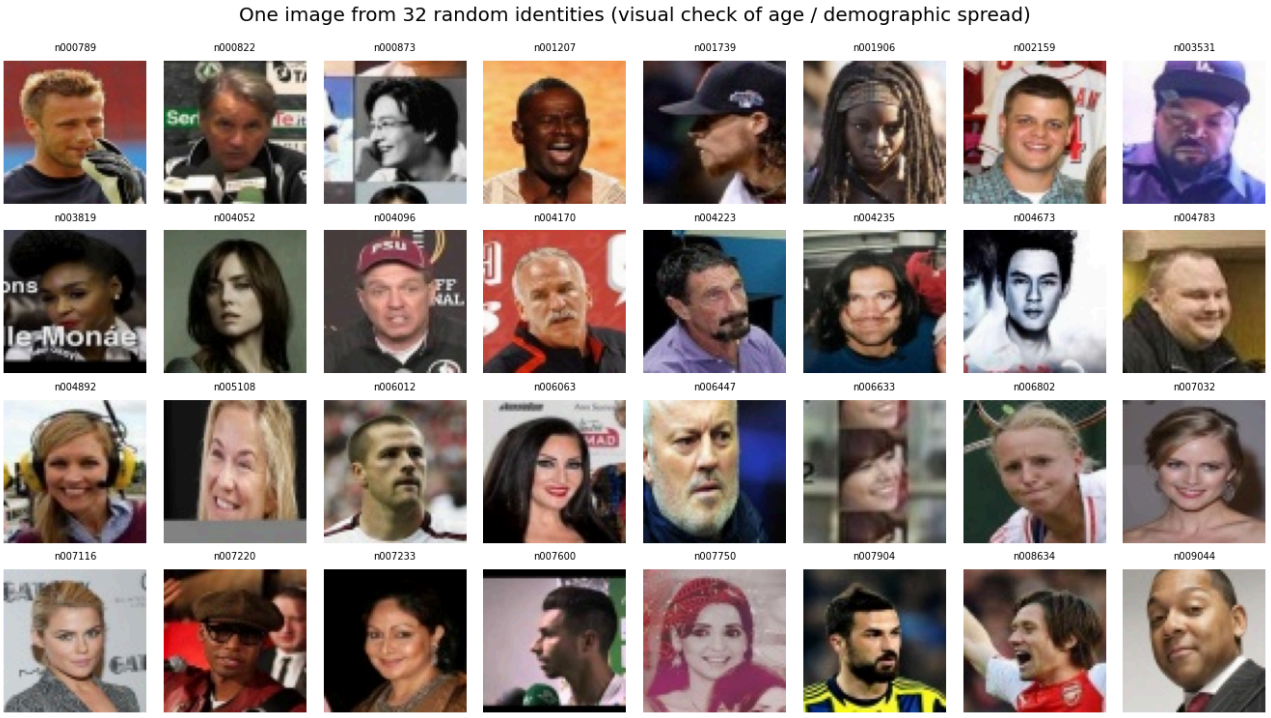


Figure 10. One image from each of 32 random identities. Captions are folder names (n + number), not real names.

The dataset contains in-the-wild photos (sports, stage, red carpet, interviews) with a visible mix of ages, genders, skin tones, hair styles and accessories such as hats and glasses. Age and demographics are **not labelled** in the dataset, and we did not try to infer them, so we cannot give exact percentages. We note this as a limitation: if the final system is used on a very different group of people, accuracy should be tested separately.

3.6 Duplicates

Type	Groups found	Where
Exact (identical file)	1	Inside one person's folder
Near copy (perceptual hash)	8	Inside one person's folder

All duplicate groups are inside a single identity, which is harmless (for example two consecutive video frames). We found **no** picture filed under two different people, which would indicate a labelling mistake.

4. Verification data

The verification folder holds 69,101 images and has no sub-folders, so there is nothing to split. Pair lists (which two images to compare) were not in the Drive folder at the time of this run, so we could only check the images themselves.

Measure	Result (6,910 images checked)

Corrupted	0
Greyscale	67 (1.0%)
Too dark	455 (6.6%)
Too bright	42 (0.6%)
Low contrast	20 (0.3%)
Size / format	Same as training: 64x64 JPEG RGB

Verification images look like training images and have the same quality profile, so a model trained on the training data can be evaluated on them without any change in image style.

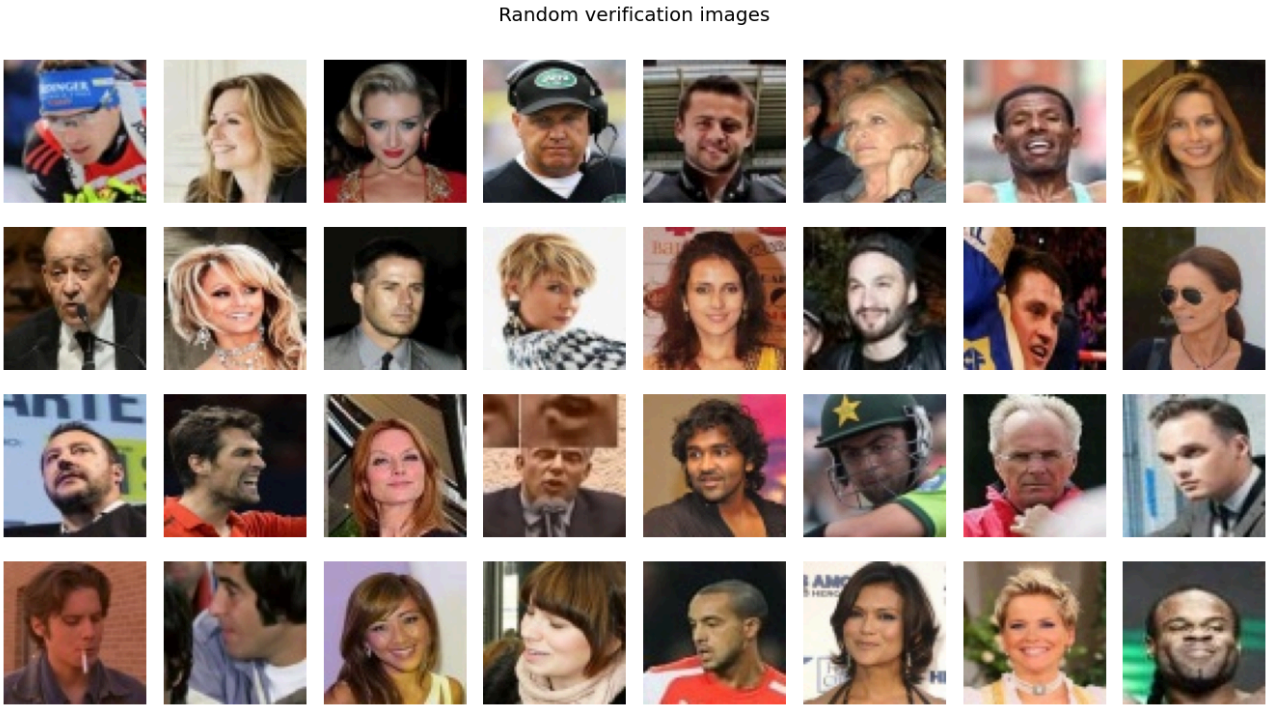


Figure 11. Random verification images.

5. Data leakage

Data leakage means that an image used to test the model was also used to train it. The model then "remembers" the answer, and the score looks better than it truly is. We compared images by their hashes (an identical file always gives an identical hash).

Compared groups	Identical images	Near copies (incl. identical)	Verdict
train vs val	0	0	Clean
train vs test	0	0	Clean
val vs test	0	0	Clean
train vs verification	73	74	**Leakage**
val vs verification	1	1	**Leakage**
test vs verification	0	0	Clean

The three classification splits do not share a single picture. The 4,000 people are shared by design (a closed-set classifier must be tested on people it knows); only the photos differ.

The verification folder, however, contains 73 images identical to training images (1.06% of the 6,910 verification images checked) and one identical to a validation image. Because only about 10% of the training images were scanned in this run, the real count is higher. A rough extrapolation suggests several hundred affected verification images in total.

> **Why this matters**
> If we train a metric-learning model on the training data and then test it on verification images that are copies of training images, those pairs become trivially easy. The AUC and the decision threshold would be too optimistic.

6. Recommendations

#	Action	Priority	Who decides
1	Run the notebook once on a zip copy of the dataset (SCAN_FRACTION = 1.0) to get exact leakage counts and a full quality scan	High	Data team
2	Remove every verification image whose hash matches a training or validation image (or remove the training copy)	High	Team agreement
3	Use verification_data for evaluation only; train on train_data only	High	All
4	Raise the blur threshold from 50 to about 450 to 700 after checking the grid; apply it to training data only	Medium	Data team
5	Keep dark images (they are real, usable faces) but use brightness / contrast augmentation	Medium	Model team
6	Keep greyscale images; add colour jitter or random greyscale augmentation	Low	Model team
7	Keep the given train / val / test split. Do not filter validation or test data	High	All
8	Obtain or create verification pair lists with same / different labels before computing AUC	High	Team lead

7. Limitations

(1) Only 10% of images were scanned, so counts are estimates. (2) Head pose and expression come from a landmark model on 64x64 images and are approximate. (3) Age and demographics are not labelled and were not inferred. (4) The pair files for verification were not available. (5) “Blurry”, “dark” and “bright” thresholds are starting values chosen by us and should be confirmed by the team.

Glossary

Term	Simple explanation
Identity / class	One person; the model learns to tell identities apart
Closed-set	Test people are the same people seen in training
Verification	Decide whether two photos show the same person
Hash (MD5)	Short fingerprint of a file; identical files have identical fingerprints
Perceptual hash	Fingerprint of how an image looks; survives resizing and re-saving
Laplacian variance	Sharpness score; low means blurry

Gini coefficient	0 = every person has the same number of images; 1 = very uneven
Yaw / pitch	Head turn left-right / tilt up-down in degrees
Data leakage	Test images that the model has already seen during training
AUC	Score of how well a verification model separates same from different pairs

Generated from the project notebook outputs (EDA_report.html, EDA_summary.md and CSV tables in the team Drive folder).